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Unit – 1 : Session – 8 : SLO – 2 | Complexity — Time and Space Trade-off

■ I. Fill in the Blanks

1. Time — the amount of time an algorithm takes.
2. Space — the amount of memory an algorithm uses.
3. time and space — the trade-off balances the two.
4. time — more memory (e.g. caching) can reduce time complexity.
5. slowly (longer) — using less space may cost more time.
6. memoization (caching) — storing precomputed values to save time.
7. operations (elementary steps) — how time complexity is measured.
8. input — space complexity includes auxiliary space plus input space.
9. space (stack) — recursion uses call-stack memory, unlike iteration.
10. constant — $O(1)$ space uses a fixed amount of memory.
11. trade-off — sorting algorithms show clear time/space trade-offs.
12. iteratively — iterative binary search is $O(\log n)$ time, $O(1)$ space.
13. computational (algorithmic) — both are forms of computational complexity.
14. computations (subproblems) — DP stores results to avoid recomputation.
15. balance (trade-off) — optimization seeks the best time–space balance.